Archive-grade scene preservation and derived object reconstruction: a combined closeout

Nested Cinema — *Vera's Not Alone*: UoS Vitrine scene pipeline and the DreamLab object sidecar

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Abstract

This report closes an engagement that joins two systems around one artwork. UoS Vitrine is a free, fully local pipeline that turns photographs and video of a physical space into a 3D Gaussian splat, packaged with its source material, camera solution, software record and checksums so the reconstruction can be treated as a defensible preservation object. The DreamLab object sidecar is a separate repository that consumes a sealed Vitrine run and derives per-object textured assets placed back into the scene, with strict separation of observed evidence from generated proposal. The test bed is *Nested Cinema — Vera's Not Alone*, Dr Pavel Prokopic's immersive installation at the University of Salford, MediaCityUK.

The validated laptop scene run registered 222 of 222 images, trained one million Gaussians in 60.4 min on an RTX 3060 Laptop GPU, and reached 25.72 dB PSNR and 0.8167 SSIM on 28 held-out views at 1.94 GB peak VRAM. An RTX 5090 reproduction of the same recipe reached 25.06 dB and 0.8211 SSIM in 2.4 min — matching laptop quality roughly 25 times faster. A workstation-archive profile (1600-pixel crop from 4096-pixel sources) suffered mass opacity collapse and is quarantined as an unvalidated negative result.

The object sidecar runs end-to-end on the sealed package. On the live installation it carved six object candidates, four well-supported (table, radio, two speakers) from ≥15 views each. On the gallery golden set every eval metric computes against 33 ground-truth object clouds; carve localises correctly but over-merges where one class tiles the scene. The reconstruction lane bake-off is partly open: Lane A (TRELLIS.2) is clean-licensed and staged; Lane B (ReconViaGen/VGGT) is research-only pending a commercial checkpoint and an unreproduced pinned environment. The two systems remain separate in code, licence and preservation responsibility, exchanging data through a versioned, checksummed file contract.

1 Context and the two systems

Site-based immersive work is hard to preserve because its meaning is spread across physical construction, media content, spatial arrangement and time. A splat alone is insufficient: it does not identify its source images, camera positions, processing environment or inferred regions. *Nested Cinema — Vera's Not Alone* makes the problem concrete. The installation premiered at Salford's MediaCity campus in June 2023 as the first public example of Prokopic's Nested Cinema practice-as-research, and the wider project secured a £298,000 AHRC Catalyst Award in March 2026. The captured subject is a room-set inside a white gallery: newspaper-clad partitions, vintage radio and reel-to-reel equipment, domestic furniture, mirrors and moving screens. Fine printed text tests sharpness; screens and reflections violate the static Lambertian assumptions that make reconstruction tractable; once de-installed, the work survives through records rather than continued access.

Table 1: Comparative architecture. Complementary, deliberately separate.

Capability	UoS Vitrine	DreamLab
Primary outcome	preservation package	structured scene
Core scene	gsplat MCMC PLY	LichtFeld / alt path
Hardware	6 GB laptop first	multi-GPU appliance
Segmentation	external	SAM3.1 pipeline
Object recon	external	TRELLIS.2 + backends
Fixity	SHA-256 package	run lineage
Licence	MIT	GPL + model terms

Two systems address adjacent parts of this problem and are deliberately kept separate. **UoS Vitrine (preservation)** asks whether a temporary installation can be captured, measured and packaged locally on modest hardware. It produces the authoritative archive master: originals with EXIF, COLMAP poses, an unquantised full-spherical-harmonic PLY, and a SHA-256 manifest. **The DreamLab object sidecar (derivation)** asks how that sealed package can yield per-object, correctly-placed, high-quality-textured assets without ever writing inside the archive. It is a separate repository with its own virtual environment behind a subprocess boundary, so no model dependency enters the MIT-licensed UoS tree.

Their capabilities are complementary rather than overlapping (Table 1). UoS Vitrine is strongest as a small, archive-first scene pipeline that runs on a 6 GB laptop; the DreamLab stack is strongest at segmentation, object reconstruction and engine-ready assembly. The object sidecar is the disciplined subset of that stack needed to close this engagement: object derivation from a sealed package, nothing more.

The split is a domain boundary, not just a packaging one. Three bounded contexts carry distinct languages: scene preservation (*capture, ingest, sfm, train, package, verify*), object derivation (*identify, rectify, carve, seed, generate, select, place, finish*), and reporting (*evidence, observed/withheld, provenance*). Two invariants run through both systems. First, the run-directory is the unit of exchange and the sidecar only ever writes alongside it, never inside. Second, observed surfaces, metrics and texels are

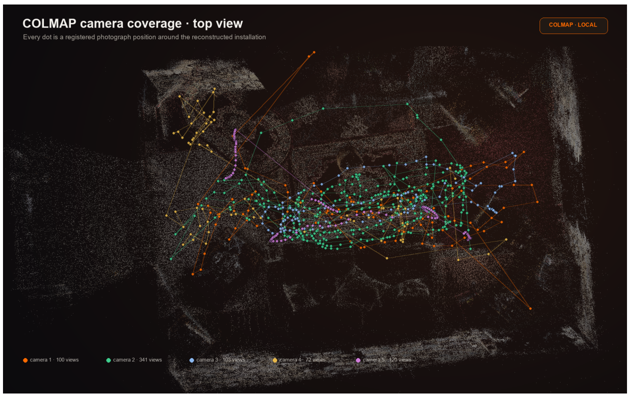

Figure 1: COLMAP camera coverage over the room-set: the 222 registered stills and video frames solved as two camera models. Source: UoS Vitrine repository (G. Watts).

never averaged with withheld or generated ones — real frames constrain, generated content proposes, and every asset carries a lineage record of its sources, checkpoints and scores.

2 Scene reconstruction results

UoS Vitrine runs as independently repeatable stages (ingest, SfM, training, evaluation, export, mesh derivation, packaging, verification), each writing a compact JSON report so expensive stages can be inspected or repeated in isolation. The same algorithm serves every hardware tier; profiles change only measured quantities (resolution, crop size, Gaussian cap, iterations). Mixed stills and video are handled with separate camera models (Figure 1): applying one set of intrinsics across both groups can silently warp the reconstruction. Training uses gsplat's MCMC densification under a hard population cap, random high-resolution crops to keep full-resolution texture in reach without rendering whole 4K frames per step, and an objective of $0.8\,L_1 + 0.2\,(1 - \mathrm{SSIM})$ because pixel-wise losses stay tolerant of blur — during development a blurred image held 10.8 dB PSNR while SSIM collapsed to 0.013.

2.1 Laptop and workstation measurements

On the target laptop (RTX 3060 Laptop, 6 GB) the validated run staged 222 images, registered all of them, and recovered two camera models and 111,203 sparse points. Training completed 15,000 iterations in 60.4 min, ending at one million Gaussians with full degree-three spherical harmonics, 1.94 GB peak VRAM, and held-out metrics of 25.72 dB PSNR / 0.8167 SSIM. VRAM was never the binding constraint; elapsed time was. The RTX 5090 control reused the proven recipe (768-pixel crop, 2048-pixel source, one-million cap, 15k iterations) and reached 25.06 dB / 0.8211 SSIM in 2.4 min at 1.75 GB, matching laptop quality about 25 times faster.

Two measurements explain why time, not memory, binds the laptop. Rasterisation cost tracks the number of Gaussians projected into the camera frustum, not the total population (Table 2): an interior camera typically observes about a third of the room, so the middle row is the practical basis for laptop estimates. The RTX 5090 reduces a comparable step to 12–14 ms, at which point kernel-launch overhead flattens the frustum–time relationship. The same six profiles span both tiers by changing only measured quantities (Table 3); the workstation-archive row records the investigated configuration, not an approved preset.

Table 2: Laptop frustum occupancy dominates iteration time (1.5M Gaussians, 768^2, SH 3).

Gaussians in view	Time / iteration
94%	500 ms
39%	222 ms
15%	95 ms

Table 3: Profiles. Same algorithm; only measured quantities change.

Profile	Src	Crop	Cap	Iters
laptop-draft	1600	512	400k	7k
laptop-standard	2048	768	1.0M	15k
laptop-archive	2560	768	1.5M	30k
workstation-draft	2048	800	1.0M	7k
workstation-standard	3200	1280	3.0M	20k
workstation-archive	4096	1600	6.0M	30k

2.2 Archive-profile opacity collapse

Our strongest tested negative result isolates crop/source scale as the decisive variable (Table 4). Both healthy runs use 768/2048; both collapsed runs use 1600/4096. Reducing the cap from six million to one million did not restore the live population, and correcting the position-learning-rate horizon did not either. In the large-crop runs the fraction of Gaussians above the live-opacity threshold falls catastrophically: one random 1600-pixel crop from a 4096-pixel source does not revisit each Gaussian often enough for reconstruction gradients to counter global opacity and scale regularisation, so MCMC relocates dead splats from a shrinking live pool. The empirical conclusion that 1600/4096 is unsafe in the current trainer is definitive; the mechanism remains a testable hypothesis. Live exported population therefore belongs beside PSNR and SSIM — a nominal six-million cap says little if 98.7% of the population is effectively transparent.

A matched-view diagnostic on held-out still `IMG_6351` (Figure 2) shows both splat renders soften sofa fabric, newspaper print and equipment detail that stay sharp in the source. The detail exists in the stills but does not survive reconstruction; the indicated next test is a stills-only baseline before reintroducing only the video frames that close a genuine coverage gap. The 5090 control is the honest working workstation result: it reproduces laptop quality about 25 times faster but does not yet exceed it.

2.3 Energy and cost

The Windows runs sample GPU power through `nvidia-smi` and integrate it over training, costed at the recorded £0.2703/kWh tariff (Table 5). The control trains for about a third of a penny; the collapsed safe-cap run costs nearly four times as much to produce a worse result. These figures cover sampled GPU power, not whole-system wall energy, so they are valid for relative trainer comparison on this workstation but are not the full institutional cost. A wall-socket meter is the preferred boundary for the planned sustainability integration.

Figure 2: Matched-view comparison on the newspaper partition: source photograph (left) against the splat reconstruction (right) from the same COLMAP camera. The printed text and clip edges stay sharp in the source and soften in the render — the detail is present in the stills but does not survive reconstruction. Source: UoS Vitrine repository (G. Watts).

Table 4: RTX 5090 controlled matrix. Crop/source scale, not cap, tracks the collapse.

Run	Crop/src	Cap	Live g.	PSNR
3060 baseline	768/2048	1.0M	555k (55%)	25.72
5090 control	768/2048	1.0M	528k (53%)	25.06
5090 candidate Q1	1024/3072	1.5M	940k (63%)	24.71
5090 safe-cap	1600/4096	1.0M	2.2k (0.2%)	17.49
5090 six-million	1600/4096	6.0M	80k (1.3%)	14.31

Table 5: Sampled-GPU energy and cost per 5090 run.

Run	Time	Energy	Cost
Control 768/2048	2.4 min	0.013 kWh	£0.0035
Q1 1024/3072	3.0 min	0.018 kWh	£0.0049
Safe-cap 1600/4096	7.6 min	0.049 kWh	£0.0132

3 Object pipeline: design and method

It consumes a Vitrine run read-only and emits `objects/objects.json` (schema `vitrine/object/1`) plus per-object GLBs and a composed scene. Nine stages run under one evidence-vs-proposal discipline.

Identify runs GroundingDINO text prompts into SAM3.1 promptable segmentation per frame, degrading to a fixed concept list rather than blocking. **Rectify** enforces an explicit image-domain contract: the sidecar persists its own undistorted images and adjusted intrinsics (because UoS never persists undistorted images), replicating `undistort.py` maths under golden tests that require pixel-identical rectification for the same camera. Every artefact records which pixel domain (distorted or rectified) it lives in.

Carve is the load-bearing correctness fix. A 2D mask votes for 3D Gaussians only through per-pixel front-depth-band gating (foreground / background / *abstain*), with min-view thresholds, 3D connected-component cleanup and robust p5–p95 extents. The naive all-along-ray voting of the earlier DreamLab `mask_projector` was rejected in adversarial review because it labels occluded Gaussians behind the object; carve is golden-tested against the gallery ground-truth object PLYs. **Seed** scores crops and views by silhouette area, sharpness, centrality and edge clearance: Lane A takes the single best matted crop, Lane B the top 8–16 diverse views. Splat renders are never used as conditioning.

Generate runs two lanes under one harness (§7). **Select** renders every candidate from all recovered real cameras and scores silhouette IoU, masked perceptual similarity, depth consistency on observed regions and mesh sanity, reporting observed and withheld surfaces separately. **Place** is an owner hard requirement: it initialises translation and scale from the carve centroid and extents, then refines full SE(3)+scale by silhouette and photometric alignment against all real masks and frames. It is symmetry-aware and fits scene-up (EXIF gravity, support-plane fit, or the LiDAR GLB cross-check on nested-cinema-04) rather than assuming $+Y$. The composed scene rendered from real cameras is the acceptance proof, and the manifest records per-axis observability, residuals and confidence.

Finish is the second hard requirement: UV unwrap (xatlas/Blender), then projective bake of real rectified frames onto observed UV islands with per-texel best-view selection by angle, sharpness and occlusion, and seam blending. Generative fill touches only unseen texels, recorded in a per-texel provenance mask, followed by albedo delighting. Real frames constrain; generated pixels propose, always flagged. **Manifest** writes one lineage record per object: sha256 of every asset *and* every checkpoint (repo, commit, hash, licence class), stage scores, coverage estimate, orientation observability and VRAM peaks. No stage may silently substitute generated data where evidence exists; a missing stage output degrades the record's status field, never fakes it.

Figure 3: From one photograph to an asset: the evidence-scored seed crop (left) and the Lane-A TRELLIS.2 reconstruction rendered from a fresh view (right), for radio, table and speaker. The generated geometry carries the crop's colour and structure; unobserved faces are inferred.

4 Live results: Nested Cinema 04

Run `nc04-object-pass1` was the sidecar's first contact with the real scene, on an RTX 6000 Ada, wall-time 352 s. The COLMAP model holds 736 registered images across five OPENCV cameras and a 2.0M-Gaussian splat; 79 frames were selected by stratified sampling. Camera 5, a 4K walkthrough of 120 registered frames, contributes nothing because the package ships only the `.MOV` and no frames were extracted; video decode is out of scope for pass 1, so this is recorded as an explicit gap. Rectify dominated wall-time (227 s of 352 s) through HEIC decode and full-resolution undistort of the 3200-class frames.

Six object candidates were carved (Table 6). Four are well-supported: the table (most-supported, 21 views), the vintage radio, and two speakers that `carve_components` correctly separated as distinct 3D instances from one

Table 6: nc04 object candidates. Four well-carved, two marginal.

Label	Gauss.	conf	views	verdict
table	38,383	0.80	21	good (best)
radio #0	17,276	0.87	16	good
armchair chair	19,055	1.00	2	weak
speaker #0	6,754	0.80	15	good
speaker #1	3,202	0.95	15	good (2nd inst.)
radio #1	6,286	0.87	16	fair (fragment?)

Figure 4: Evidence-scored seed crops from the live installation.

`speaker` mask set. Two are problematic: `armchair chair` rests on only two views (a confidence of 1.00 is unreliable at that count, and the label is phrase-merged), and `radio#1` is a smaller second component that may be a genuine second unit or a fragment. Seed crops confirm real installation objects: vintage radios, cassette decks, a record-player case, a table. Figure 4 shows three of the evidence-scored seed crops.

4.1 Carve metrics on the golden set

On the gallery golden set the full object path ran over 121 frames and scored each carved object against 33 ground-truth object clouds. Every eval metric computes — the primary validation goal. Carve localises the right region but does not yet separate instances on an adversarial scene: the gallery is wall-to-wall same-class paintings, so per-label mask unions cover most of each frame and carve votes a large fraction of the splat foreground. Three "objects" of 1.5–2.4M Gaussians each overlap ground-truth `object_035` with high reproj-IoU (0.84–0.89) but only 3D-F1 ≈ 0.35; smaller 3D components recover individual objects better (best 3D-F1 0.519) but many are fragments (F1 < 0.15). This contrasts with nested-cinema-04, where discrete, spatially separated objects carved cleanly. The method is sound where objects are separated and degrades where a class tiles the scene; the biggest available fix is per-instance masks rather than per-label unions.

4.2 Per-instance carve: pass 1 to pass 2

Pass 2 selected all 736 registered frames (including Camera 5's walkthrough, now extracted) and switched carve to per-instance masks. Object support rose sharply: `radio#0` went from 17,276 Gaussians over 16 views to 118,070 over 331, `table#0` to 39,155 over 250, `speaker#0` to 52,964 over 242. Nine candidates passed and three were rejected on the min-view floor (`armchair`, `gramophone`, `mirror`, all `insufficient_views`) rather than kept as weak detections — the pass-1 "weak pair" problem is now handled by a rejected list, not a caveat. The cost is wall-time: full-frame rectify and identify push the pass to 4662 s, dominated by rectify (2297 s) and identify (1891 s).

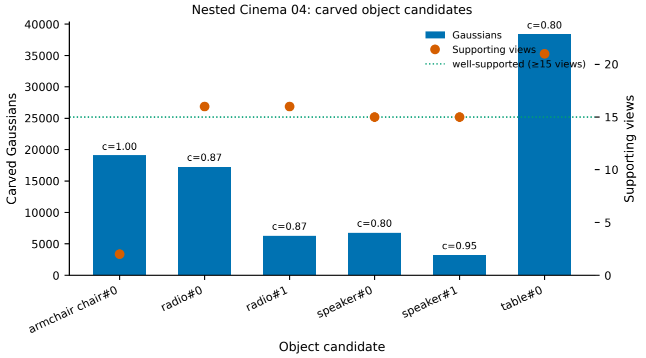

Figure 5: Carved candidates: Gaussian count (bars), supporting views (markers) and carve confidence. The two below the 15-view line (armchair at 2 views, radio#1 fragment) are the weak pair.

4.3 Generation bake-off

Both lanes now produce assets that fit the ship target (Figure 6, §7). **Lane A** (TRELLIS.2, single matted crop at 1024^3) generated 12 of 12 candidates — radio, table and speaker × four seeds — with no failures. Reserved-VRAM peaks were 9.1 GB (radio), 5.3 GB (table) and 7.7 GB (speaker), and per-seed runtimes ran 37 s to 120 s. **Lane B** (ReconViaGen, textured GLB) produced a single-view radio at 15.0 GB and a multi-view radio (top-8 IoU-ranked views) at 16.6 GB; both textured GLBs are ≈17 MB. Every variant clears the 32 GB ceiling, Lane A with roughly 3.5× headroom.

A first `composed_scene.glb` exists for the radio, placing the Lane-A and Lane-B assets into the scene at solved transforms under an explicit basis contract. Each object carries provenance classes in `composed_scene_objects.json`: `geometry_class=generated`, `material_class=generated`, `placement_class=carve-init`, and `derivative_class=composed-derivative`, so the placement is marked as carve-centroid initialised, not yet silhouette-refined.

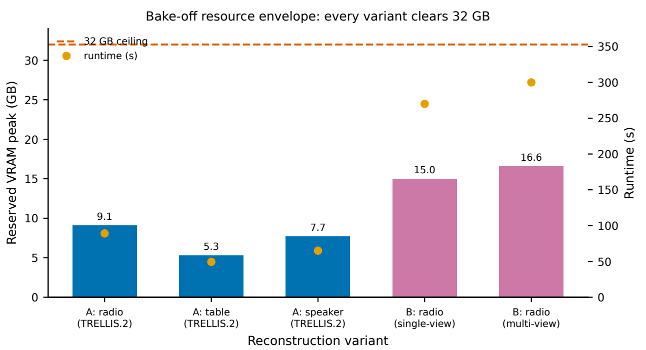

Figure 6: Bake-off resource envelope: reserved-VRAM peak per variant against the 32 GB ceiling, with runtime. Lane-A peaks are the max across four seeds; Lane-B are the textured single/multi-view runs.

Quality ranking across lanes is deferred: the nc04 numeric silhouette scores are not yet reliable (the scale bracket was only just fixed via the manifest `scene_scale`, and refined scoring is running), so visual contact sheets are the current evidence and no numeric lane winner is claimed.

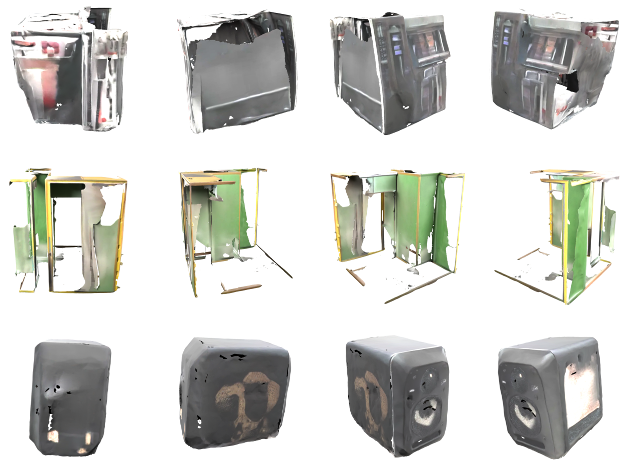

Figure 7: Lane-A TRELLIS.2 objects, four turntable views each (radio, table, speaker; best seed by mesh detail). Real generated 3D assets with baked texture; carving and single-crop generation leave visible holes on unobserved backs.

The ranking chart lands when refined scoring completes (§5.1).

5 Evaluation

Evaluation is observed-vs-withheld throughout. A held-out azimuth sector (25% of views on the gallery set) is never merged into observed metrics, so backside quality cannot be inflated by front-face coverage. Three metric families run: 3D F1 and IoU by voxel-occupancy overlap of carved Gaussian centres against the best-matching ground-truth cloud (voxel = 1% scene scale = 0.123); reproj-IoU, a coarse point-splat silhouette reprojected into each camera against the SAM mask; and mesh sanity (component count, watertightness, non-manifold edges) run on the reference mesh as a baseline for scoring generated meshes later. The ground-truth `full_scene.glb` is itself messy (11,583 verts, 356 components, not watertight, 654 non-manifold edges), which is a useful yardstick.

Two limits qualify the current numbers, both recorded plainly. The reproj-IoU is a lower bound because there is no true rasteriser: a coarse splat silhouette muddies both the absolute IoU and the observed/withheld gap, which at this coverage is small and noisy (sometimes withheld exceeds observed). A real gsplat rasteriser would sharpen both. And the golden set may encode the old pipeline's assumptions, so it is used for golden regression tests only; a blinded fresh capture with withheld angular sectors is the true backside benchmark.

Fairness rules for the lane bake-off (from adversarial review) are common mask and frame-selection inputs, identical finishing and eval, and both fixed-candidate-count and fixed-GPU-time comparisons with seed variance reported — not just winners. Candidate selection runs at $512^3/1024^3$ and the winner is regenerated at ship quality.

5.1 Charts

One plot shows the over-merge (Figure 9). Every marker is a carved object scored against its best-matching ground-truth cloud; area is the carved Gaussian count. Three over-merged blobs (≥1M Gaussians) sit top-left: high

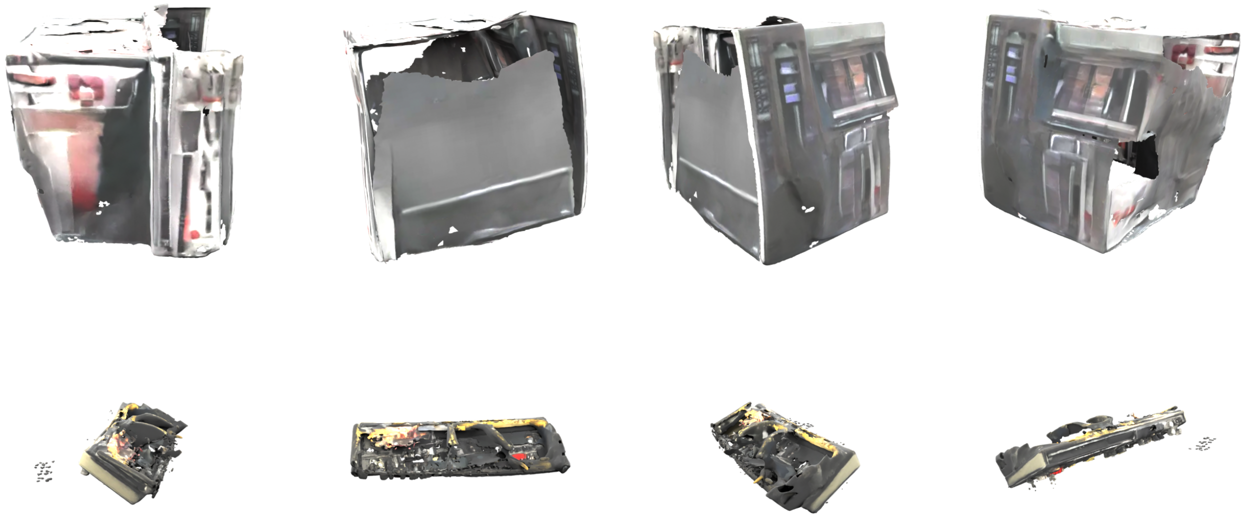

Figure 8: Bake-off, visual: the radio reconstructed by Lane A (TRELLIS.2 single crop, top) and Lane B (ReconViaGen multi-view, bottom), same four viewpoints. Lane A is a solid, closed housing with softer detail; Lane B recovers more panel structure but strews floaters and thin shells. With numeric silhouette scoring still settling, this is the current bake-off evidence.

reproj-IoU because they cover the frame, low 3D-F1 because they are not one object, while the smaller clean components reach higher F1 to the right. The lane bake-off chart fills once Lane-A/B runs land.

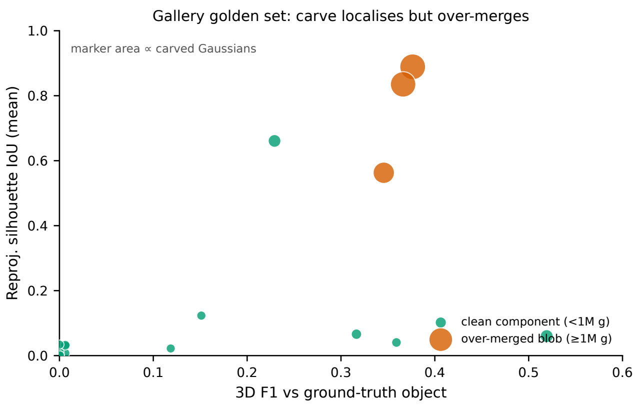

Figure 9: Carve localises but over-merges where a class tiles the scene. Marker area $\propto$ carved Gaussians.

Resource cost across the bake-off is settled (Figure 6, §4.3): every Lane-A and Lane-B variant clears the 32 GB ceiling. Quality ranking is not. The nc04 numeric silhouette scores are not yet reliable, because the object-to-scene scale bracket was only just fixed by reading `scene_scale` from the manifest and refined scoring is still running. Until it completes, visual contact sheets are the evidence of record and no numeric lane winner is claimed. **[TODO: Quality-ranking chart: per-object silhouette IoU / masked-DINO across lanes, once refined scoring lands.]**

6 Interface contract and the UoS PR

Data crosses the boundary as files, not imports. UoS exports a versioned handoff containing the authoritative scene-PLY hash, the registered staged images, the COLMAP text model, capture metadata, the coordinate convention, the parent archive-manifest hash and stable object requests. The sidecar validates the handoff and bypasses its own ingest, SfM and scene retraining. The original full-scene PLY stays byte-identical and authoritative; a cleaned derivative `scene_cleaned.ply` may be produced by subtracting the exact union of accepted object selections, preserving selection indices, thresholds and before/after counts, with no inferred background fill by default. Cleaned scene and generated objects are engine derivatives linked to the master, not replacements for it.

The UoS-side change is kept minimal and additive. `package.py` gains an optional `objects` manifest key and copies `runs/<name>/objects/` into `archive/derivatives/objects/`. `cli.py` gains a `vitrine objects` subcommand that invokes the sidecar as a subprocess. `serve.py` gains an optional read-only objects panel. No scene-mesh wiring ships in v1. The integration seam already exists and stands up: the sidecar venv imports uos-vitrine cleanly and the CLI exposes `doctor/ingest/sfm/train/package/objects/evaluate/verify/run`, with `objects` already shelling out to the sidecar.

Validation surfaced one real cross-platform defect. `vitrine verify nested-cinema-04` re-checksums the manifest but rejects every file: the package was authored on Windows and `manifest.json` stores `originals\<name>.jpg` with backslash separators, which the POSIX path-safety guard flags as an unsafe path. The packager should emit POSIX separators, or verify should normalise them before the safety check. This is filed for the UoS PR. Two pull requests are open against the UoS repository: PR #1 (the sidecar hook: the `objects` subcommand and manifest key, three commits, 62 tests) and PR #2 (an additive `composed_scene` manifest field, stacked on PR #1).

7 Licence and deployment envelope

The shipped configuration must fit the UoS target: Windows 11, RTX 5090, 32 GB VRAM, sm_120 (Blackwell). Development and validation ran on 48 GB RTX 6000 Ada cards. Per-stage VRAM peaks are to be measured under an enforced 32 GB ceiling before any config is declared 5090-fit, and a real Windows/5090 acceptance run is the release gate. If that run cannot be executed from the development container, it is recorded as an explicit release blocker rather than hand-waved. **[TODO: Windows/5090 acceptance run on real hardware: this remains the release gate.]** The measured evidence so far is under budget on every path. The object pass's instrumented stage (identify) peaked at 6.6 GB (Figure 10); generation reserved at most 9.1 GB on Lane A and 16.6 GB on Lane B (Figure 6). All clear the 32 GB ceiling, Lane A with roughly 3.5× headroom, so the acceptance run is expected to confirm rather than discover the envelope.

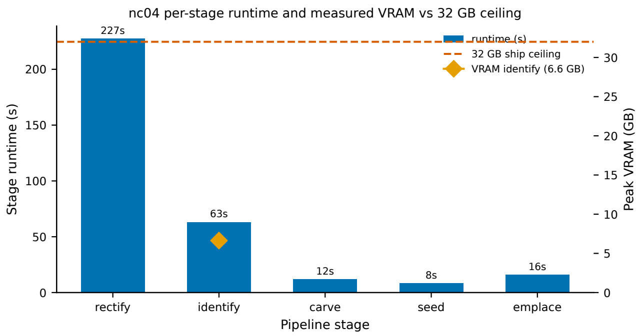

Figure 10: nc04 per-stage runtime with the measured VRAM peak against the 32 GB ship ceiling. Only `identify` was VRAM-instrumented in pass 1; the others are marked as unmeasured rather than assumed.

Licence gating decides what can ship. Table 7 reproduces the audited findings. The decisive point: ReconViaGen v0.5 does not load the Facebook VGGT checkpoint — it loads `Stable-X/vggt-object-v0-1`, a VGGT-architecture derivative with *no declared licence*, so a commercial deployment must treat it as encumbered by Meta's VGGT CC-BY-NC terms until clarified. The plain VGGT arm loads `facebook/VGGT-1B` (CC-BY-NC, research/eval only); its commercial checkpoint is gated behind a manual application form and was not accepted on the owner's behalf. Consequently **Lane B is research/eval-only for this closeout** unless the commercial checkpoint is obtained. Lane A (TRELLIS.2-4B, MIT) and everything else in the finishing path are clean. Every checkpoint used is recorded in lineage: repo, commit, hash and licence class.

Two environment blockers keep Lane B unreproduced here. ReconViaGen pins conda py3.10 + torch 2.4/cu121

Table 7: Model licence gate (audited 2026-08-28).

Component	Licence	Gate
ReconViaGen v0.5 code	MIT	clean
microsoft/TRELLIS.2-4B	MIT	clean
Stable-X/trellis-vggt-v0-2	MIT	clean
Stable-X/vggt-object-v0-1	none declared	treat as NC
facebook/VGGT-1B	CC-BY-NC-4.0	eval only
facebook/VGGT-1B-Commercial	gated form	not accepted
Hunyuan3D-2.1	Tencent custom	review before ship
SAM3.1 checkpoint (local)	unverified	verify before ship

with five source-compiled CUDA extensions and cp310 prebuilt wheels (flash-attn, kaolin, xformers), none installable on the container's nix py3.12; and it ships only a Gradio demo, so a headless run needs a custom driver written after the environment is solved. The recommended route is a `uv`-managed standalone py3.10 in the workspace (mechanism confirmed; the Docker-via-host-socket route is a documented mount trap). The VGGT arm is fully operational: a smoke test inferred six gallery frames in 0.94 s at 8.1 GB peak. Lane A TRELLIS.2-4B weights (16 GB) are staged on disk by symlink from a read-only mount, so no re-download was needed.

8 Limitations and negative results

Recording failure is part of the preservation method. The clearest scene-side negative result is the archive-profile opacity collapse (§2): the 1600/4096 workstation-archive profile drives 98.7–99.8% of the Gaussian population transparent and is quarantined as unvalidated. Omitting it would create the false impression that six million Gaussians are inherently superior; reporting the full matrix constrains future claims and keeps throughput capability distinct from reconstruction quality.

On the object side, carve over-merges where one class tiles the scene. On the gallery golden set, per-label mask unions collapse wall-to-wall paintings into a few multi-million-Gaussian blobs that overlap one ground-truth object; individual instances are recovered only as smaller, sometimes fragmentary components. The fix is per-instance masks fed to carve separately, not per-label unions. Orientation remains the weakest emplacement axis: placement solves translation and scale from the carve subset reliably, but full orientation must be recovered from silhouettes and is marked `needs_review` when under-constrained — low-confidence placements must not be presented as validated assets, and generated backsides and occluded surfaces are labelled inferred.

Further limits, stated plainly: a splat interpolates views, it does not measure surfaces, and COLMAP recovers arbitrary scale unless an external measurement establishes world units. Mirrors, screens, glass and polished metal are represented by observed pixels, not physical optics; moving screens are constitutive to the artwork and a static splat cannot preserve their temporal sequence. The live nc04 pass leaves Camera 5's 4K walkthrough (120 frames) unused because only the `.MOV` shipped — the densest trajectory is currently absent. GroundingDINO phrase-merging pollutes labels ("armchair chair", "radio typewriter"); it is cosmetic for carve but needs one prompt per detect call. And the reproj-IoU metric is a lower bound until a real gsplat rasteriser replaces the coarse point-splat silhouette.

9 Recommendations and handover

Scene pipeline. Quarantine the 1600/4096 workstation-archive profile and keep the 25.06 dB RTX 5090 control as the verified workstation baseline. Do not promote the Q1 candidate — it stayed healthy but scored lower. Future tuning should vary one axis at a time (post-densification refinement, multi-crop batches, opacity regularisation) with live-opacity percentage monitored as an acceptance metric, rejecting runs below a declared health threshold. Run a stills-only 5090 baseline from the 72 archival masters before requesting any reshoot, and complete the formal newspaper-legibility and matched-view Luma comparisons using saved cameras rather than a single subjective orbit.

Object pipeline. The biggest available fix is per-instance masks: feed each detection box's SAM mask into carve separately so tiling classes do not collapse into one blob. Add a real gsplat rasteriser for silhouette, PSNR and SSIM to sharpen the observed/withheld gap. Complete the lane bake-off once Lane B's environment is stood up via the `uv` py3.10 route, and resolve the Stable-X checkpoint licence before any commercial Lane B ship. Extract Camera 5's walkthrough frames to add the densest trajectory, and normalise GroundingDINO prompts to one per detect call.

Integration and handover. Land the minimal additive UoS PR (§6), including the Windows→POSIX path normalisation in the packager/verify. Seal the preferred scene package with originals, COLMAP text model, training report, derivatives and limitations under a stable run identifier before creating any object or engine derivatives. Preserve stable IDs across masks, crops, selections, reconstructed assets and placements, and disable cloud fallbacks for preservation runs. The two repositories stay separate in code, licence, product identity and preservation responsibility; the file contract is the boundary that keeps them so.

Both UoS PRs are open (sidecar hook #1, composed-scene field #2). The remaining gates are the Windows/5090 acceptance run, the refined-scoring lane winner regenerated at ship quality, PR review and merge, and the sealed-package hashes. **[TODO: Tick these off at final handover.]**

Appendix A: environment snapshot

Table 8: Validated environments. Scene pipeline (UoS) and object sidecar (DreamLab).

Component	Validated value
Scene Python / Torch	3.11 venv; Torch 2.11, CUDA 12.8
gsplat	1.5.3
Laptop GPU	RTX 3060 Laptop, 6 GB, compute 8.6
Workstation GPU	RTX 5090 class, 32 GB (ship target sm_120)
COLMAP	Docker image
Master splat	Full-SH 3DGS PLY; SHA-256 manifest
Sidecar Python / Torch	nix py3.12; Torch 2.4.0 cu121 (Lane B venv)
Dev GPUs	RTX 6000 Ada (sm_89) + A6000 (sm_86), 48 GB
TRELLIS.2-4B weights	16 GB, symlinked from read-only mount
VGGT smoke	6 frames, 0.94 s, 8.1 GB peak (facebook/VGGT-1B, eval only)
Sidecar test suite	112 passed, 0 failed (not-slow)

Appendix B: preservation package schema

```
archive/
  originals/     untouched camera photographs and video
  sfm/           cameras.txt, images.txt, points3D.txt, database.db
  model/         scene.ply, full spherical harmonics
  derivatives/   web splat, mesh, previews, objects/ (sidecar output)
  manifest.json  versions, settings, metrics, SHA-256 fixity
  README.md      plain-language context and verification instructions
```

Manifest schema identifier `vitrine/preservation-package/1`; object records `vitrine/object/1`. Verification detects corruption or unintended change; it does not replace a backup and replication policy.

Appendix C: ReconViaGen environment blockers

- **B1**: ReconViaGen `setup.sh` pins conda py3.10 + torch 2.4.0/cu121 with cp310 prebuilt wheels (flash-attn 2.7.0.post2, kaolin, xformers) and five source-compiled CUDA extensions (nvdiffrast, nvdiffrec, CuMesh, FlexGEMM, o-voxel); none install on the nix py3.12 venv. Recommended route: `uv`-managed standalone py3.10 (mechanism confirmed).
- **B2**: v0.5 ships only a Gradio demo (`app_v05.py`), no headless CLI; an automated run needs a custom driver around `TrellisHybridPipeline` after B1.
- **B3**: 256 MB tmpfs caches hit ENOSPC; resolved by redirecting HF/pip/TMP caches to the workspace disk.